



Final Standards Track: ERC

ERC-5192: Minimal Soulbound NFTs

Minimal interface for soulbinding EIP-721 NFTs

Authors Tim Daubenschütz (@TimDaub), Anders (@0xanders)

Created 2022-07-01

Requires EIP-165, EIP-721

Table of Contents

- Abstract
- Motivation
- Specification
 - Contract Interface
- Rationale
- Backwards Compatibility
- Security Considerations
- Copyright

Abstract

This standard is an extension of EIP-721. It proposes a minimal interface to make tokens soulbound using the feature detection functionality of EIP-165. A soulbound token is a non-fungible token bound to a single account.

Motivation

The Ethereum community has expressed a need for non-transferrable, non-fungible, and socially-priced tokens similar to World of Warcraft's soulbound items. But the lack of a token standard leads many developers to simply throw errors upon a user's invocation of transfer functionalities. Over the long term, this will lead to fragmentation and less composability.

In this document, we outline a minimal addition to EIP-721 that allows wallet implementers to check for a token contract's permanent (non-)transferability using EIP-165.

Specification

The keywords "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY" and "OPTIONAL" in this document are to be interpreted as described in RFC 2119.

Contract Interface

A token with a `uint256 tokenId` may be bound to a receiving account with `function locked(...)` returning `true`. In this case, all EIP-721 functions of the contract that transfer the token from one account to another must throw.

```
// SPDX-License-Identifier: CC0-1.0
pragma solidity ^0.8.0;

interface IERC5192 {
    /// @notice Emitted when the locking status is changed to locked.
    /// @dev If a token is minted and the status is locked, this event should be emitted.
    /// @param tokenId The identifier for a token.
    event Locked(uint256 tokenId);

    /// @notice Emitted when the locking status is changed to unlocked.
    /// @dev If a token is minted and the status is unlocked, this event should be emitted.
    /// @param tokenId The identifier for a token.
    event Unlocked(uint256 tokenId);

    /// @notice Returns the locking status of an Soulbound Token
    /// @dev SBTs assigned to zero address are considered invalid, and queries about them do throw.
    /// @param tokenId The identifier for an SBT.
    function locked(uint256 tokenId) external view returns (bool);
}
```

To aid recognition that an EIP-721 token implements "soulbinding" via this EIP upon calling EIP-721's `function supportsInterface(bytes4 interfaceID) external view returns (bool)` with `interfaceID=0xb45a3c0e`, a contract implementing this EIP must return `true`.

Rationale

The above model is the simplest possible path towards a canonical interface for Soulbound tokens. It reflects upon the numerous Soulbound token implementations that simply revert upon transfers.

Backwards Compatibility

This proposal is fully backward compatible with EIP-721.

Security Considerations

There are no security considerations related directly to the implementation of this standard.

Copyright

Copyright and related rights waived via CC0.

Citation

Please cite this document as:

Tim Daubenschütz (@TimDaub), Anders (@0xanders), "ERC-5192: Minimal Soulbound NFTs," *Ethereum Improvement Proposals*, no. 5192, July 2022. Available: <https://eips.ethereum.org/EIPS/eip-5192>.